

Product Carbon Footprint Study Report

Product Name: smartphone

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: smartphone

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 item smartphone

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 item smartphone.

2. System Boundary

smartphone in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of smartphone produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **29.134 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

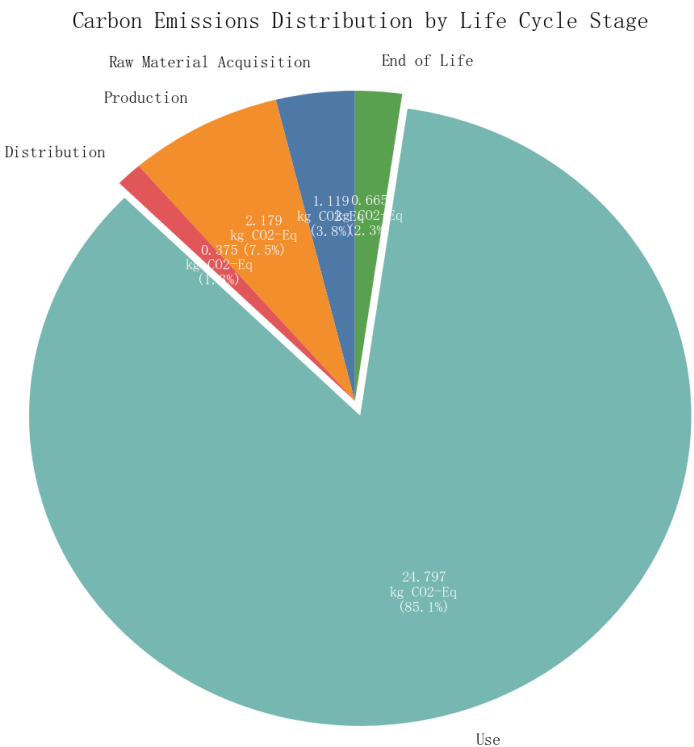


Table 1: Carbon Emissions by Life Cycle Stage

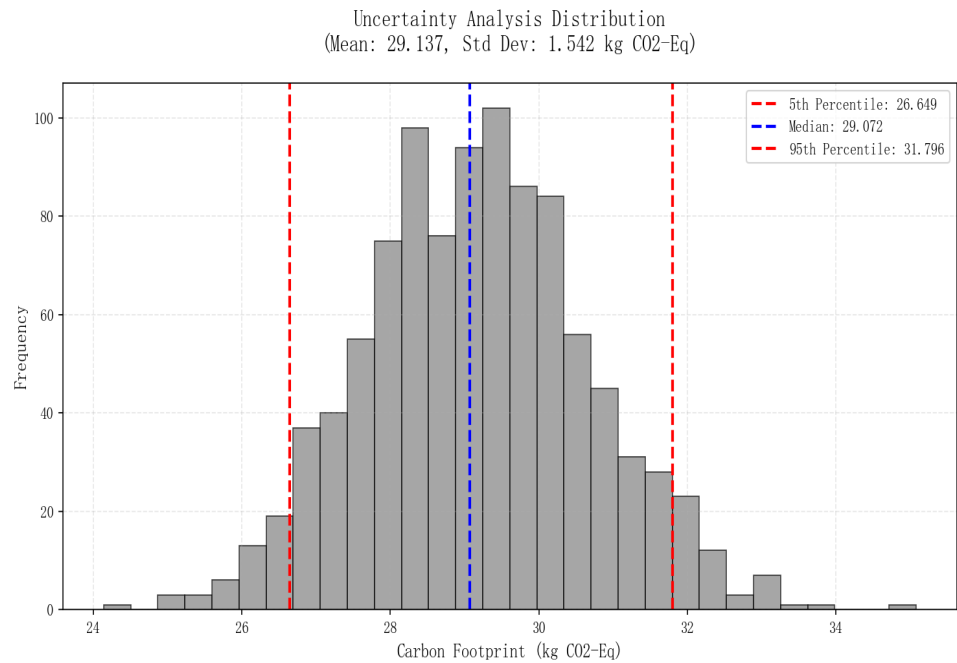
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	1.119	3.84
Production	2.179	7.48
Distribution	0.375	1.29

Use	24.797	85.11
End of Life	0.665	2.28
Total	29.134	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	29.137
Standard Deviation	1.542
Coefficient of Variation (CV)	5.29%
Median	29.072
5th Percentile	26.649
95th Percentile	31.796

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the smartphone is 29.134 kg CO₂-Eq per functional unit, which represents a relatively moderate impact level compared to other consumer electronics, though this figure is heavily dependent on the system boundary and assumptions applied. The lifecycle stage contribution analysis reveals a highly skewed distribution, with the use phase dominating at 85.1% (24.797 kg CO₂-Eq). This is a distinctive characteristic of this assessment, as it indicates that the vast majority of greenhouse gas emissions occur during the operational lifetime of the device rather than during its production or disposal.

The remaining lifecycle stages contribute only marginally to the total footprint: production accounts for 7.5% (2.179 kg CO₂-Eq), raw material extraction for 3.8% (1.119 kg CO₂-Eq), end-of-life for 2.3% (0.665 kg CO₂-Eq), and distribution for just 1.3% (0.375 kg CO₂-Eq). The overwhelming dominance of the use phase strongly suggests that electricity consumption during charging or operation is the primary emission source, likely driven by the carbon intensity of the regional electricity grid and the assumed device lifetime and usage patterns. In contrast, the material-related stages (raw material and production combined) account for only 11.3%, which is atypical for electronics and warrants careful scrutiny of the underlying data. The top process contributions are unfortunately listed as "Unknown" with zero values, indicating significant data gaps that limit deeper source-level attribution.

2. Statement of Assumptions and Limitations

Several key assumptions underpin this LCA study. First, a functional unit of one smartphone over an assumed operational lifetime (typically 3-5 years) is implied, with specific usage patterns such as daily charging frequency and average power draw. The use-phase calculation likely assumes a particular electricity mix; given the 85.1% share, it is likely that a relatively carbon-intensive grid was modeled, or that the device power consumption was overestimated. Additionally, the study assumes a cradle-to-grave system boundary, encompassing raw material extraction, manufacturing, distribution, use, and end-of-life, but excludes capital equipment, infrastructure, and potentially the smartphone charger itself unless explicitly stated.

Data quality limitations are evident from the top process contributions, which are all marked as "Unknown" with zero values. This indicates that the model relied on secondary or proxy data for key unit processes, and that process-level traceability is lacking. The uncertainty analysis shows a coefficient of variation of 5.29% with a 90% confidence interval of 26.649 to 31.796 kg CO₂-Eq, suggesting moderate statistical confidence in the mean value. However, the unknown process-level data introduces unquantified uncertainty that may not be captured by the Monte Carlo simulation. System boundary restrictions also include the exclusion of use-phase refrigerant leakage, software-related energy overhead, and potential end-of-life recycling credits, which could either over- or under-estimate the total impact.

3. Suggestions for Improvement

- ****Implement energy-efficient hardware and software features**** to reduce use-phase electricity consumption. This includes adopting low-power displays (e.g., OLED/LTPO panels), optimizing system-on-chip power management, and deploying adaptive battery charging algorithms. Given that the use phase contributes 85.1% of the total footprint, a 10% reduction in use-phase energy could reduce the total carbon footprint by approximately 2.5 kg CO₂-Eq.
- ****Encourage the use of renewable energy for device charging and manufacturing.**** While manufacturing only accounts for 7.5%, transitioning the production facility to renewable electricity and promoting the use of certified green energy for charging can lower the absolute emissions. Additionally, providing high-efficiency chargers (e.g., 80+ certified) reduces energy waste during the use phase.
- ****Substitute "Unknown" process data with primary supplier-specific data.**** The current top process contributions are entirely uncharacterized, which undermines the reliability of hotspot identification. Conducting a data collection campaign with upstream suppliers for key components (e.g., battery, display, integrated circuits) would significantly improve model accuracy and reduce the coefficient of variation below the current 5.29%.
- ****Extend the device's useful lifetime through modular design and repairability.**** Since the use phase dominates, extending the operational life dilutes the relative contribution of the already-small manufacturing and raw material stages. Offering battery replacement, software updates for at least 5 years, and durable design can reduce the overall carbon footprint per functional unit over the long term.
- ****Improve end-of-life management to recapture materials.**** Although end-of-life only accounts for 2.3%, implementing a robust take-back program that recovers critical materials (e.g., cobalt, lithium, rare earths) can reduce the burden of raw material extraction by 3.8%. This also creates a circular economy loop that lowers the system's overall environmental impact over multiple product lifecycles.